Cognizant Programming Questions for Freshers

Python Programming Questions (30)

1. Write a program to swap two numbers without using a third variable.
2. Write a program to check if a given number is prime or not.
3. Write a program to find the factorial of a number using recursion.
4. Write a program to generate the Fibonacci series up to N terms.
5. Write a program to check if a string is a palindrome or not.
6. Write a program to reverse a string without using built-in functions.
7. Write a program to count the number of vowels and consonants in a string.
8. Write a program to remove all duplicates from a string while preserving order.
9. Write a program to find the first non-repeating character in a string.
10. Write a program to check if two strings are anagrams of each other.
11. Write a program to find the second largest element in a list.
12. Write a program to merge two sorted lists into one sorted list.
13. Write a program to remove all duplicates from a list.
14. Write a program to find the common elements between two lists.
15. Write a program to rotate a list by N positions.
16. Write a program to count the frequency of each character in a string using a dictionary.
17. Write a program to sort a dictionary by its values.
18. Write a program to find the intersection of two sets.
19. Write a program to find the union of two sets.
20. Write a program to check if a key exists in a dictionary.
21. Write a function that takes a list of numbers and returns only the even numbers.
22. Write a program using lambda to sort a list of tuples by the second element.
23. Write a function that accepts variable arguments (*args) and returns their sum.
24. Write a program using map() to convert a list of temperatures from Celsius to Fahrenheit.
25. Write a program using filter() to find all numbers divisible by 3 in a list.
26. Create a class 'Student' with attributes name, roll_no, and marks. Add a method to calculate grade.
27. Write a program demonstrating inheritance with a base class 'Animal' and derived class 'Dog'.
28. Write a program to demonstrate method overriding in Python.

29. Create a class with private attributes and demonstrate encapsulation using getters and setters.

30. Write a program to demonstrate multiple inheritance in Python.

SQL Programming Questions (30)

1. Create a table 'Employee' with columns: EmpID (INT, PK), Name (VARCHAR), Age (INT), Salary (DECIMAL), DeptID (INT).

2. Insert 5 records into the 'Employee' table with sample data.

3. Write a query to update the salary of an employee by 10% where EmpID = 101.

4. Write a query to delete employees whose age is less than 18.

5. Write a query to add a new column 'Email' to the 'Employee' table.

6. Write a query to select all employees whose salary is between 30000 and 50000.

7. Write a query to find all employees whose name starts with 'A'.

8. Write a query to list employees ordered by salary in descending order.

9. Write a query to find employees whose age is greater than the average age of all employees.

10. Write a query to find employees not in department IDs 10, 20, or 30.

11. Write a query to find the total number of employees in each department.

12. Write a query to find the average salary per department.

13. Write a query to find the maximum and minimum salary in the company.

14. Write a query to find departments having more than 5 employees.

15. Write a query to find the total salary paid by each department.

16. Write a query using INNER JOIN to display employee name and department name (assume 'Department' table exists).

17. Write a query using LEFT JOIN to show all employees and their departments (even if no department assigned).

18. Write a query to find employees who are not assigned to any department (using LEFT JOIN).

19. Write a query to join three tables: Employee, Department, and Project to show employee details with their project.

20. Write a query to find employees working in the same department using self-join.

21. Write a query to find the second highest salary from the Employee table.

22. Write a query to find the third highest salary from the Employee table.

23. Write a query to find employees whose salary is greater than the average salary of their department.

24. Write a query to find the Nth highest salary (generalized query).

25. Write a query to find employees earning more than the average salary of the company.

26. Write a query to find duplicate records in the Employee table based on Name and Age.

27. Write a query to delete duplicate rows from a table, keeping only one instance.

28. Write a query to find employees who joined in the last 6 months (assume JoinDate column *exists*).

29. Write a query to create a view that shows employee name, department, and salary.

30. Write a query to find the top 3 highest-paid employees in each department.

 # Cognizant Interview – Fundamental Questions for Freshers

Python Fundamental Interview Questions (50)

1. What is Python? What are its key features?
2. What is the difference between Python 2 and Python 3?
3. Explain the importance of indentation in Python.
4. What are the different data types in Python? Give examples.
5. What is the difference between mutable and immutable data types?
6. How do you take user input in Python?
7. What are the rules for naming variables in Python?
8. Explain the difference between '/' and '//' operators.
9. What is the use of the '**' operator?
10. How do you swap two variables in Python?
11. How do you reverse a string in Python?
12. What is string slicing? Explain with examples.
13. How do you check if a string is a palindrome?
14. What is the difference between '==' and 'is' in Python?
15. How do you convert a string to lowercase and uppercase?
16. How do you check if two strings are anagrams?
17. How do you find the first non-repeating character in a string?
18. How do you count the frequency of each character in a string?
19. How do you remove duplicates from a string while preserving order?
20. How do you reverse words in a sentence?
21. What is the difference between list and tuple?
22. How do you add an element to a list?
23. How do you remove duplicates from a list?
24. What is list comprehension? Give an example.
25. How do you sort a list in descending order?
26. How do you find the second largest element in a list?
27. How do you merge two sorted lists into one sorted list?
28. How do you find common elements between two lists?
29. How do you rotate a list by N positions?
30. How do you find the index of an element in a list?
31. Why are tuples immutable? What are their advantages?

32. How do you convert a list to a tuple?
33. What is the difference between set and list?
34. How do you find the union of two sets?
35. How do you find the intersection of two sets?
36. How do you find the difference between two sets?
37. How do you check if a set is a subset of another set?
38. What is a dictionary in Python?
39. How do you access a value in a dictionary?
40. How do you add a new key-value pair to a dictionary?
41. How do you remove a key from a dictionary?
42. What happens if you try to access a non-existent key?
43. How do you sort a dictionary by its values?
44. How do you merge two dictionaries?
45. How do you count the frequency of elements using a dictionary?
46. What is the difference between a function and a method?
47. What are *args and **kwargs?
48. What is a lambda function? Give an example.
49. What is the difference between local and global variables?
50. What is a recursive function? Give an example.
51. Write a function that takes a list of numbers and returns only the even numbers.
52. Write a program using lambda to sort a list of tuples by the second element.
53. Write a function that accepts variable arguments (*args) and returns their sum.
54. Write a program using map() to convert Celsius to Fahrenheit.
55. Write a program using filter() to find all numbers divisible by 3.
56. What is a class and object in Python?
57. Explain the __init__ method.
58. What is inheritance? Give an example.
59. What is polymorphism? Explain with an example.
60. What is encapsulation? How is it achieved in Python?
61. What is abstraction? How is it different from encapsulation?
62. What is the difference between instance variable and class variable?
63. What is method overriding?
64. What is multiple inheritance? Does Python support it?
65. What is the 'self' keyword in Python?
66. Create a class 'Student' with attributes name, roll_no, and marks. Add a method to calculate grade.
67. Write a program demonstrating inheritance with a base class 'Animal' and derived class 'Dog'.
68. Create a class with private attributes and demonstrate encapsulation using getters and setters.
69. What is exception handling? Why is it important?
70. Explain try, except, finally blocks with an example.
71. What is the difference between 'except' and 'except Exception'?
72. How do you raise an exception in Python?
73. What is a custom exception? How do you create one?
74. Write a program to handle division by zero error.
75. What is a KeyError? How do you handle it?
76. How do you open a file in Python?
77. What are the different modes to open a file?
78. How do you read all lines from a file?

79. How do you write to a file in Python?
80. How do you count the number of words in a file?
81. What is a module in Python?
82. How do you import a module?
83. What is the difference between 'import module' and 'from module import *'?
84. What is a package in Python?
85. What is the use of the 'if __name__ == "__main__"' statement?
86. Write a program to check if a given number is prime or not.
87. Write a program to find the factorial of a number using recursion.
88. Write a program to generate the Fibonacci series up to N terms.
89. Write a program to find duplicates in an array.
90. Write a program to move all zeros to the end of an array.
91. Write a program to find the missing number in an array of 1 to N.
92. Write a program to find the majority element in an array.
93. Write a program to implement a stack using a list.
94. Write a program to implement a queue using a list.
95. Write a program to check if parentheses are balanced using a stack.
96. Write a program to find the maximum sum subarray (Kadane's algorithm).
97. Write a program to find the longest substring without repeating characters.
98. Write a program to find all anagrams of a pattern in a string.
99. Write a program to implement LRU cache.
100. Write a program to find the kth largest element in an array.

SQL Fundamental Interview Questions (50)

Basic Concepts

 # Cognizant Interview – Top 100 Most Asked SQL Questions

1. What is SQL? What is it used for?

2. What is the difference between SQL and MySQL?

3. What are the different types of SQL commands?

4. Explain DDL, DML, DCL, and TCL with examples.

5. What is a primary key? Can it be NULL?

6. What is a foreign key? How is it different from a primary key?

7. What is a unique key? How is it different from a primary key?

8. What is a composite key?

9. What is a candidate key?

10. What are constraints in SQL? Name a few.

11. How do you create a table in SQL?

12. How do you add a new column to an existing table?

13. How do you delete a column from a table?
14. What is the difference between DROP, TRUNCATE, and DELETE?
15. How do you rename a table in SQL?
16. Write a query to create an Employee table with EmpID, Name, Age, Salary, DeptID.
17. Write a query to insert 5 records into the Employee table.
18. Write a query to update the salary of an employee by 10%.
19. Write a query to delete employees whose age is less than 18.
20. Write a query to add a new column 'Email' to the Employee table.
21. What is the use of the SELECT statement?
22. What is the difference between WHERE and HAVING clause?
23. How do you use the LIKE operator? Give examples.
24. What is the use of the DISTINCT keyword?
25. How do you filter records using IN operator?
26. Write a query to select all employees whose salary is between 30000 and 50000.
27. Write a query to find all employees whose name starts with 'A'.
28. Write a query to find employees not in department IDs 10, 20, or 30.
29. Write a query to find employees whose name contains ' Kumar'.
30. Write a query to select employees with NULL in a specific column.
31. How do you sort results in ascending and descending order?
32. Can you use ORDER BY with multiple columns? Explain.
33. What is the default sorting order in SQL?
34. How do you limit the number of rows returned?
35. What is the difference between LIMIT and TOP?
36. Write a query to list employees ordered by salary in descending order.
37. Write a query to find the top 5 highest-paid employees.
38. Write a query to sort employees by department and then by salary.
39. What are aggregate functions? Name a few.
40. What is the difference between COUNT(*) and COUNT(column_name)?
41. How do you find the average of a column?
42. How do you find the maximum and minimum value in a column?
43. What is the use of GROUP BY clause?

44. Write a query to find the total number of employees in each department.

45. Write a query to find the average salary per department.

46. Write a query to find the maximum and minimum salary in the company.

47. Write a query to find departments having more than 5 employees.

48. Write a query to find the total salary paid by each department.

49. What is a JOIN in SQL?

50. What is the difference between INNER JOIN and OUTER JOIN?

51. Explain LEFT JOIN with an example.

52. Explain RIGHT JOIN with an example.

53. What is a FULL OUTER JOIN?

54. What is a self-join? When is it used?

55. What is a cross join?

56. How many tables can you join in a single query?

57. What is the difference between UNION and UNION ALL?

58. What is the difference between JOIN and UNION?

59. Write a query using INNER JOIN to display employee name and department name.

60. Write a query using LEFT JOIN to show all employees and their departments.

61. Write a query to find employees who are not assigned to any department.

62. Write a query to join three tables: Employee, Department, and Project.

63. Write a query to find employees working in the same department using self-join.

64. What is a subquery? Give an example.

65. What is a correlated subquery?

66. Write a query to find the second highest salary from the Employee table.

67. Write a query to find the third highest salary from the Employee table.

68. Write a query to find the Nth highest salary (generalized query).

69. Write a query to find employees whose salary is greater than the average salary.

70. Write a query to find employees whose salary is greater than the average salary of their department.

71. Write a query to find duplicate records in a table.

72. Write a query to delete duplicate rows from a table, keeping only one instance.

73. Write a query to find employees who joined in the last 6 months.

74. Write a query to find the top 3 highest-paid employees in each department.

75. What is a view in SQL? Why is it used?

76. What is an index? Why is it used?

77. What is a stored procedure?

78. What is a trigger in SQL?

79. Write a query to create a view that shows employee name, department, and salary.

80. Write a query to create an index on the Salary column.

81. What is normalization? Explain 1NF, 2NF, 3NF.

82. What is denormalization? When is it used?

83. What is BCNF (Boyce-Codd Normal Form)?

84. What is a transaction in SQL?

85. What are ACID properties in a transaction?

86. Write a query to find employees earning more than their managers.

87. Write a query to find the department with the highest average salary.

88. Write a query to find employees who have the same salary.

89. Write a query to find the cumulative salary by department.

90. Write a query to rank employees by salary within each department.

91. Write a query to find consecutive numbers appearing at least three times.

92. Write a query to find the second highest salary without using LIMIT.

93. Write a query to pivot rows to columns (basic pivot).

94. Write a query to find the running total of salaries.

95. What is a window function? Give an example.

96. What is the difference between ROW_NUMBER(), RANK(), and DENSE_RANK()?

97. Write a query to find the top N salaries using window functions.

98. Write a query to calculate moving average of salaries.

99. Write a query to find gaps in a sequence of numbers.

100. Write a query to implement pagination using OFFSET and FETCH.

Cognizant Interview – Cloud, ML & DS Fundamental Questions

Cloud Fundamentals (50)

Cloud Basics

1. What is cloud computing? What are its main benefits?

2. Explain the three main service models of cloud computing: IaaS, PaaS, SaaS.

3. What is the difference between public, private, and hybrid cloud?

4. What are the major cloud service providers? Name at least three.

5. What is meant by 'on-demand' in cloud computing?

Compute & Storage

6. What is a virtual machine (VM)? How is it different from a container?

7. What is serverless computing? Give an example.

8. What is object storage? How is it different from block storage?

9. What is a CDN (Content Delivery Network)? Why is it used?

10. What is auto-scaling in cloud computing?

Networking & Security

11. What is a VPC (Virtual Private Cloud)?

12. What is the difference between security groups and network ACLs?

13. What is IAM (Identity and Access Management)? Why is it important?

14. What is multi-factor authentication (MFA) in cloud?

15. What is encryption at rest and encryption in transit?

Deployment & Management

16. What is cloud migration? What are common strategies?

17. What is Infrastructure as Code (IaC)? Name a tool.

18. What is the difference between horizontal and vertical scaling?

19. What is a load balancer? Why is it used?

20. What is cloud monitoring? Name a few metrics to monitor.

Cost & Optimization

21. What is the pay-as-you-go model in cloud?

22. What are reserved instances vs on-demand instances?

23. What is cloud cost optimization? Name a few techniques.

24. What is a cloud region and availability zone?

25. What is disaster recovery in cloud? Name a strategy.

Containers & Orchestration

26. What is Docker? What problem does it solve?

27. What is Kubernetes? What is it used for?

28. What is the difference between a Docker image and a container?

29. What is a container registry?

30. What is a pod in Kubernetes?

Cloud-Native Concepts

31. What is a microservices architecture?

32. What is an API gateway?

33. What is event-driven architecture in cloud?

34. What is a managed service? Give examples.

35. What is the difference between stateful and stateless applications?

Security & Compliance

36. What is the principle of least privilege in IAM?

37. What is a security group in AWS or equivalent in other clouds?

38. What is DDoS protection in cloud?

39. What is compliance in cloud computing? Name a standard.

40. What is shared responsibility model in cloud security?

Advanced Cloud Concepts

41. What is cloud-native development?

42. What is a cloud function or serverless function?

43. What is edge computing? How is it related to cloud?

44. What is multi-cloud strategy? What are its pros and cons?

45. What is cloud bursting? When is it used?

Practical Scenarios

46. How would you design a highly available web application in cloud?

47. What steps would you take to secure a cloud database?

48. How do you monitor and alert on cloud resource usage?

49. What would you consider when choosing a cloud region for deployment?

50. How would you reduce cloud costs for a startup?

Machine Learning & Data Science Fundamentals (50)

ML Basics

1. What is machine learning? How is it different from traditional programming?

2. What are the three main types of machine learning?

3. What is supervised learning? Give two examples.

4. What is unsupervised learning? Give two examples.

5. What is reinforcement learning? Give an example.

Data & Preprocessing

6. What is a dataset? What are features and labels?

7. What is the difference between training data and testing data?

8. Why is data preprocessing important in ML?

9. What is missing value imputation? Name a few techniques.

10. What is feature scaling? Why is it needed?

11. What is one-hot encoding? When is it used?

12. What is label encoding? How is it different from one-hot encoding?

13. What is train-test split? Why is it done?

14. What is cross-validation? Why is it useful?

15. What is data leakage? How can you avoid it?

Model Evaluation

16. What is accuracy? When is it not a good metric?

17. What is precision and recall? When do you prioritize each?

18. What is F1-score? Why is it used?

19. What is a confusion matrix? Explain TP, TN, FP, FN.

20. What is ROC-AUC? What does it measure?

21. What is overfitting? How can you detect it?

22. What is underfitting? How is it different from overfitting?

23. What is the bias-variance tradeoff?

24. What is a learning curve? What does it show?

25. What is the difference between validation set and test set?

Algorithms – Supervised Learning

26. What is linear regression? When is it used?

27. What is logistic regression? Is it used for regression or classification?

28. What is a decision tree? What are its advantages and disadvantages?

29. What is a random forest? How is it different from a single decision tree?

30. What is support vector machine (SVM)? When is it useful?

31. What is k-nearest neighbors (KNN)? How does it work?

32. What is gradient boosting? Name a popular library.

33. What is XGBoost? Why is it popular?

Algorithms – Unsupervised Learning

34. What is clustering? Give an example algorithm.

35. What is k-means clustering? How does it work?

36. What is hierarchical clustering?

37. What is dimensionality reduction? Why is it used?

38. What is PCA (Principal Component Analysis)? What does it do?

Model Training & Tuning

39. What is hyperparameter tuning? Why is it important?

40. What is grid search? How does it work?

41. What is random search? How is it different from grid search?

42. What is regularization? Name L1 and L2 regularization.

43. What is dropout in neural networks?

44. What is early stopping? Why is it used?

Deep Learning Basics

45. What is a neural network? What are layers, neurons, and weights?

46. What is an activation function? Name a few (ReLU, Sigmoid, Softmax).

47. What is backpropagation? Why is it important?

48. What is a convolutional neural network (CNN)? Where is it used?

49. What is a recurrent neural network (RNN)? Where is it used?

50. What is transfer learning? Give an example.

Cognizant Interview – Top 50 Most Asked Data Structures Questions

1. What is a data structure? Why is it important?

2. What is the difference between linear and non-linear data structures?

3. What is time complexity? Explain Big O notation.

4. What is space complexity? How is it different from time complexity?

5. What is the difference between O(1), O(log n), O(n), and O(n^2)?

6. What is an array? What are its advantages and disadvantages?

7. How do you find the second largest element in an array?

8. How do you reverse an array in place?

9. How do you rotate an array by k positions?

10. How do you find duplicates in an array?

11. How do you check if two strings are anagrams?

12. How do you check if a string is a palindrome?

13. How do you find the first non-repeating character in a string?

14. How do you reverse words in a sentence?

15. What is a linked list? How is it different from an array?

16. What is a singly linked list vs doubly linked list?

17. How do you reverse a linked list?

18. How do you detect a cycle in a linked list?

19. How do you find the middle element of a linked list?

20. How do you merge two sorted linked lists?

21. What is a stack? What operations does it support?

22. How do you implement a stack using an array?

23. How do you check if parentheses are balanced using a stack?

24. How do you implement a min-stack (getMin in O(1))?

25. What is a queue? What operations does it support?

26. How do you implement a queue using an array?

27. How do you implement a queue using two stacks?

28. What is a priority queue? Where is it used?

29. What is recursion? What are its advantages and disadvantages?

30. What is base case and recursive case in recursion?

31. Write a recursive function to calculate factorial of a number.
32. Write a recursive function to calculate the Nth Fibonacci number.
33. What is linear search? What is its time complexity?
34. What is binary search? What are its prerequisites?
35. Explain bubble sort, selection sort, and insertion sort.
36. Explain merge sort and quick sort. What are their time complexities?
37. What is a hash table? How does it work?
38. What is collision in hashing? How is it resolved?
39. How do you implement a hash map in your language of choice?
40. How do you count frequency of elements using a hash map?
41. What is a binary tree? What is a binary search tree (BST)?
42. What are inorder, preorder, and postorder traversals?
43. How do you find the height of a binary tree?
44. How do you check if a binary tree is a BST?
45. What is BFS (Breadth-First Search)? How does it work?
46. What is DFS (Depth-First Search)? How does it work?
47. What is Dijkstra's algorithm? What problem does it solve?
48. What is dynamic programming? How is it different from recursion?
49. Explain the 0/1 knapsack problem.
50. How do you find the maximum sum subarray (Kadane's algorithm)?

 # Cognizant Interview – Top 20 Most Asked SDLC Questions

1. What is SDLC? Why is it important in software development?
2. Explain the different phases of SDLC.
3. What is the difference between SDLC and STLC?
4. What are the different SDLC models? Name at least four.
5. Explain the Waterfall model with its advantages and disadvantages.
6. What is the Iterative model? How is it different from Waterfall?
7. Explain the Spiral model. When is it preferred?
8. What is the V-Model (Verification and Validation model)?
9. Explain the Agile model. What are its core principles?

10. What is Scrum? How is it related to Agile?
11. What is a sprint in Agile? What happens in a sprint?
12. What is the difference between Agile and DevOps?
13. What is requirements gathering? Why is it critical?
14. What is the difference between functional and non-functional requirements?
15. What is system design? What are its inputs and outputs?
16. What is the difference between high-level design (HLD) and low-level design (LLD)?
17. What is coding/implementation phase in SDLC?
18. What is testing in SDLC? Name different types of testing.
19. What is deployment? Explain different deployment strategies.
20. What is maintenance in SDLC? What activities are involved?